



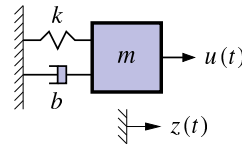
## Example of a continuous-time state-space model

- Representation of the dynamics of an  $n$ th-order system as a first-order differential equation in an  $n$ -vector called the state.  
 $\Rightarrow n$  coupled first-order equations.

- Classic example: Second-order equation of motion.

$$m\ddot{z}(t) = u(t) - b\dot{z}(t) - kz(t)$$

$$\Rightarrow \ddot{z}(t) = \frac{u(t) - b\dot{z}(t) - kz(t)}{m},$$



where  $\ddot{z}(t) = d^2z(t)/dt^2$ ,  $\dot{z}(t) = dz(t)/dt$ , etc.

- Define a (non-unique) state vector:

$$x(t) = \begin{bmatrix} z(t) \\ \dot{z}(t) \end{bmatrix}, \quad \text{so, } \dot{x}(t) = \begin{bmatrix} \dot{z}(t) \\ \ddot{z}(t) \end{bmatrix} = \begin{bmatrix} \dot{z}(t) \\ -\frac{k}{m}z(t) - \frac{b}{m}\dot{z}(t) + \frac{1}{m}u(t) \end{bmatrix}.$$



## Example in continuous-time state-space form

- So far, we have:

$$\dot{x}(t) = \begin{bmatrix} \dot{z}(t) \\ \ddot{z}(t) \end{bmatrix} = \begin{bmatrix} \dot{z}(t) \\ -\frac{k}{m}z(t) - \frac{b}{m}\dot{z}(t) + \frac{1}{m}u(t) \end{bmatrix}.$$

- We can write this as  $\dot{x}(t) = Ax(t) + Bu(t)$ , where  $A$  and  $B$  are constant matrices.

$$\dot{x}(t) = \begin{bmatrix} \dot{z}(t) \\ \ddot{z}(t) \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}}_A \begin{bmatrix} z(t) \\ \dot{z}(t) \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_B u(t).$$

- Complete the model by computing  $z(t) = Cx(t) + Du(t)$ , where  $C$  and  $D$  are constant matrices.

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \end{bmatrix}.$$



## Standard state-space model form

- Standard form for continuous-time linear state-space model:

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t)$$

$$z(t) = Cx(t) + Du(t) + v(t),$$

where  $u(t)$  is the known input,  $x(t)$  is the state,  $A$ ,  $B$ ,  $C$ ,  $D$  are constant matrices.

- Convenient way to express dynamics (matrix format great for computers).
- The first equation is called the state equation or the process equation.
  - Notice that this is the only equation that evolves over time (since it is a first-order vector ODE that integrates the right-hand side to find  $x(t)$ ).
  - So, this equation summarizes the “dynamics” of the model.
- The second equation is called the output equation or the measurement equation.
  - It is a static linear combination of variables known at time  $t$ .



## What is the system state vector?

- Standard form for continuous-time linear state-space model:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) + w(t) \\ z(t) &= Cx(t) + Du(t) + v(t).\end{aligned}$$

- We think of the system state at time  $t_0$  as a minimum set of information at  $t_0$  that, together with input  $u(t)$ ,  $t \geq t_0$ , uniquely determines system behavior for all  $t \geq t_0$ .
  - State variables provide access to what is going on *inside* the system.
  - The state has dimensions  $x(t) \in \mathbb{R}^n$ , so  $A \in \mathbb{R}^{n \times n}$  and  $w(t) \in \mathbb{R}^n$ .
- Note that in principle we can solve the state equation numerically,

$$x(t) = x(0) + \int_0^t Ax(\tau) + Bu(\tau) + w(\tau) d\tau,$$

but it will usually be more convenient to keep the equation in differential form.



## What are the output and inputs?

- Standard form for continuous-time linear state-space model:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) + w(t) \\ z(t) &= Cx(t) + Du(t) + v(t).\end{aligned}$$

- The model output is  $z(t) \in \mathbb{R}^m$ . This is the measurement we make.
- There are three input signals to the model:  $u(t)$ ,  $w(t)$ , and  $v(t)$ .
- We consider  $u(t) \in \mathbb{R}^r$  to be a (deterministic) input; that is, we assume that we know its value exactly at all times.
  - Based on the size of  $u(t)$ , we infer that  $B \in \mathbb{R}^{n \times r}$ .
  - The deterministic input forces  $x(t)$  to evolve over time in different ways, depending on its values.
  - It also influences the output via the  $D$  term.



## What are the random (stochastic) inputs?

- Standard form for continuous-time linear state-space model:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) + w(t) \\ z(t) &= Cx(t) + Du(t) + v(t).\end{aligned}$$

- The model has two random input signals:  $w(t)$ , and  $v(t)$ .
- They are random in the sense that we assume that we never know their values exactly.
  - We have already seen that  $w(t) \in \mathbb{R}^n$ ; if  $z(t) \in \mathbb{R}^m$ , then  $v(t) \in \mathbb{R}^m$  also.
- The  $w(t)$  signal is process noise. Notice that it affects the dynamics of the model by making direct changes to the evolution of  $x(t)$ .
- The  $v(t)$  signal is sensor noise. Notice that it does not affect the dynamics of the model; it affects only the measurement  $z(t)$ .
- Systems having noise inputs  $w(t)$  and  $v(t)$  are considered in detail in week 3.



## What are the matrices called?

- Standard form for continuous-time linear state-space model:

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t)$$

$$z(t) = Cx(t) + Du(t) + v(t).$$

- $A \in \mathbb{R}^{n \times n}$  is the system matrix. It models the evolution of the state in the absence of input.
- $B \in \mathbb{R}^{n \times r}$  is the input matrix. It defines how linear combinations of  $u(t)$  impact the evolution of the state.
- $C \in \mathbb{R}^{m \times n}$  is the output matrix. It defines how the output depends on linear combinations of states.
- $D \in \mathbb{R}^{m \times r}$  is the feedforward (or feedthrough) matrix. It models how the output depends on linear combinations of the input (instantaneously).
- Time-varying systems have  $A, B, C, D$  that change with time.



## Why do you need to know about them?

- It can be of vital interest to know the state of the system, but we usually cannot measure it directly.
- Instead, we *can* measure  $u(t)$  and  $z(t)$ , and although we cannot measure the random signals  $w(t)$  and  $v(t)$ , we can model some of their critical attributes.
- This enables us to make methods to estimate  $x(t)$  based on what we measure and what we model.
- Kalman filters are (in some cases optimal) estimators of  $x(t)$ .
  - The derivation and implementation of the KF depends on modeling the system in state-space form and having knowledge of the model  $A, B, C$ , and  $D$  matrices.
  - This is why we care!



## Summary

- State-space models are a compact representation of an  $n$ th-order linear system in terms of a 1st-order vector ODE.
- The standard form for linear continuous-time state-space models is:

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t)$$

$$z(t) = Cx(t) + Du(t) + v(t),$$

- You have learned the names of the equations, the names of the matrices, and the names of the signals (and what they all mean).
- You have seen one example of a model in state-space form; in the next lesson, you will be introduced to three other important general models.